Monisha Singh

Curriculum Vitae

Ph.D. Candidate, Computer Science & Engineering, Indian Institute of Technology Ropar, Punjab, India

Research Interests

I build multimodal machine learning systems that understand human social behaviour and adapt to it in real time. My doctoral work covers new datasets of engagement and attention, interpretable multimodal models of engagement and affect, and a deployed engagement and affect-aware agent evaluated with 60 participants. I want to bring this work to socially intelligent robots: systems that read people's engagement and affect robustly, decide when and how to act, explain those decisions, and are evaluated with real users in settings such as healthcare.

Areas: multimodal machine learning, social signal processing, affective computing, human-robot and human-AI interaction, foundation models, interpretable and adaptive AI.

Education

Ph.D., Computer Science & Engineering, Indian Institute of Technology Ropar | January 2022 – Present
Research area: Affective Computing, Computer Vision | Supervisor: **Dr. Abhinav Dhall** | GPA: 8.2/10

M.S., Computer Science, Northeastern University, Boston, MA, USA | GPA: 3.8/4

B.E., Computer Science & Engineering, RTM Nagpur University, Nagpur, India | 71.4%

Research Experience

Ph.D. Candidate, IIT Ropar | January 2022 – Present

- **Multimodal social understanding:** built models of engagement, attention and affect from facial behaviour, gaze, head pose and video. Trained a dyadic engagement backbone on NoXi and transferred it across domains with an ordinal regression head.
- **Foundation models:** fine-tuned video-language models (VideoLLaMA2, Qwen3-VL) and Llama-3-8B with LoRA and QLoRA for engagement and attention recognition. Quantified the signal lost when gaze, head pose and facial expression are converted into text for LLM input.
- **Adaptive, neuro-symbolic agent:** built a real-time agent that combines neural perception of engagement and affect with explicit rules for when to intervene, chooses among six strategies with Thompson sampling, explains its decisions to the user, and generates responses with a local LLM (Qwen3.5-9B).
- **Interpretable models:** graph-based mixture-of-experts and affect-integrated models of engagement, explained through attention scores and facial Action Units.
- **Deployment and evaluation with people:** deployed the real-time agent (FastAPI, React, TypeScript, OpenFace) in a three-arm between-subjects study (N = 60) and analysed the results with permutation and equivalence tests.
- **Datasets and behaviour generation:** co-created EngageNet, a large-scale engagement dataset, and DREAMS (32 participants, 781 videos). Building a triadic podcast-conversation dataset for generating listener reactions.

Publications

Journals

- [**IEEE TAC 2026**] EngageNet++: Affective Integration and Interpretable Multimodal Engagement Prediction.
  M. Singh, X. Hoque, D. Zeng, Y. Wang, K. Ikeda, S. S. Khan, A. Dhall. *IEEE Transactions on Affective Computing, accepted September 2026.*

Conferences

- [**ICMI 2026**] Interpretable Multimodal Engagement Prediction with Graph-based Mixture-of-Experts.
  M. Singh, A. Dhall. *ACM ICMI 2026, Naples, Italy.*
- [**ICPR 2024**] DREAMS: Diverse Reactions of Engagement and Attention Mind States Dataset.
  M. Singh, G. Sharma, X. Hoque, A. Dhall. *ICPR 2024, Kolkata, India.*
- [**ICMI 2023**] Do I Have Your Attention? A Large-Scale Engagement Prediction Dataset and Baselines.
  M. Singh, X. Hoque, D. Zeng, Y. Wang, K. Ikeda, A. Dhall. *ACM ICMI 2023, Paris, France.*
- [**ICMI 2023**] EmotiW 2023: Emotion Recognition in the Wild Challenge.
  A. Dhall, **M. Singh**, R. Goecke, T. Gedeon, D. Zeng, Y. Wang, K. Ikeda. *ACM ICMI 2023, Paris, France.*

Under review

- [**Journal**] Decoding DREAMS: Understanding Engagement and Attention through Video and Language Models.
  M. Singh, G. Sharma, A. Dhall. *Image and Vision Computing (Elsevier), submitted October 2026.*
- [**Conference**] An engagement and affect-aware conversational agent for adaptive tutoring, with a 60-participant user study.
  M. Singh, S. Ghosh, A. Dhall, J. Shukla. *Full paper under double-blind review at a top-tier HCI conference.*

In preparation

- A multimodal multi-party conversation dataset from podcasts for listener-reaction generation.

Work Experience

Teaching Assistant, Indian Institute of Technology Ropar

- Courses: Machine Learning, Data Structures, Operating Systems and Tinkering Lab.

Software Development Engineer, Amazon, Seattle, WA, USA

- Independently set up a new merchant capacity service and management stack for the Prime Now launch in a new realm.
- Migrated the cart client and removed Prime Now checkout's hard dependency on Fresh, improving performance and reliability.
- Cut notification service response time with a client-side cache (Guava) and built browser automation tests.

Software Development Engineer Intern, Amazon, Seattle, WA, USA

- Developed the Prime Now Shop Past Purchases page, front end and back end.

Honors and Awards

- **Best Student Paper Award**, ACM ICMI 2026, Naples, Italy.
- Fully funded scholarship to attend ACM ICMI 2023, Paris.
- Google Scholar (October 2026): 109 citations, h-index 3, i10-index 2.

Datasets Released

- **EngageNet**: a large-scale dataset for engagement prediction (ACM ICMI 2023).
- **DREAMS**: 32 participants, 781 videos, engagement and attention labels (ICPR 2024).

Collaborations

- KDDI Research, Japan: Kazushi Ikeda, Yanan Wang, Donghuo Zeng (EngageNet, EngageNet++, EmotiW 2023).
- IIIT-Delhi, Human-Machine Interaction Lab: Dr. Jainendra Shukla (tutoring-agent user study).
- IIT Kharagpur: Dr. Shreya Ghosh (tutoring-agent user study).
- American University of the Middle East, Kuwait: Dr. Shehroz S. Khan (EngageNet++).

Technical Skills

Areas: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Large Language Models (LLMs), Multimodal Learning
Languages and systems: Python, Java, C/C++, TypeScript, SQL, FastAPI, React, MongoDB, Git, LaTeX
ML frameworks and tools: PyTorch, Hugging Face Transformers, PEFT (LoRA, QLoRA), OpenFace, Ollama, Keras
Research methods: User-study design and ethics protocols, permutation and equivalence testing, mixed-effects analysis